

Lab 3: Component Modelling & Architectural Pattern Selection

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Written Justification – Group Travel Expense Simplification Engine

The Group Travel Expense Simplification Engine is designed using a component-based architecture where the system is divided into different subsystems and components. Each component is responsible for a specific task, which makes the overall system easier to understand and manage.

The system is divided into User & Access Management, Expense Management, Data Management, and Settlement & Payment subsystems. This separation helps keep related functionalities together and avoids making one component responsible for too many tasks. For example, expense-related operations are handled separately from settlement and payment operations.

The components communicate with each other through interfaces and APIs. These interfaces define how one component uses the services provided by another component. For example, the Expense Manager communicates with the Currency Conversion Service when currency conversion is required. Similarly, the Balance Calculator passes the calculated balance information to the Settlement Engine.

The Database is kept as a separate component so that important information such as user details, group information, and expense records can be stored and accessed by the required components. Authentication is also separated from the main application functions to control access to the system.

The Settlement & Payment subsystem handles the final stage of the process. The Balance Calculator determines the individual balances, the Settlement Engine generates a simplified settlement plan, and the Payment Service handles the required payment processing.

This component-based structure provides modularity and separation of responsibilities. It makes the system easier to develop, test, maintain, and modify. If a particular component needs to be changed or improved, the other components do not need to be completely redesigned. Overall, the diagram gives a clear representation of the system's major components, their responsibilities, and the way they communicate with each other.